Table 1: Clustering quality results for the ADULT dataset.

					Lloyd		Cost					$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]				Balance					
k	λ	Target	p	q	Cost	Bal.	MIQCP	SetVars	MPFC	Flow	SFC	MIQCP	SetVars	MPFC	Flow	Lloyd	MIQCP	SetVars	MPFC	Flow	SFC
2	0.01	0.49	462	941	3,685	0.41	-	3,753	**3,697**	3,698	4,062	-	-7.63	-8.99	-8.97	✗	-	✓	✓	✓	✓
2	0.10	0.45	420	941			-	3,696	**3,688**	**3,688**	3,904	-	-5.33	-5.53	-5.54	✗	-	✓	✓	✓	✓
2	0.30	0.35	310	893			-	**3,685**	**3,685**	3,686	3,756	-	-1.89	-1.89	-1.86	✓	-	✓	✓	✓	✓
5	0.01	0.49	485	988	2,370	0.29	-	2,593	**2,434**	2,447	3,310	-	-21.66	-26.47	-26.09	✗	-	✓	✓	✓	✓
5	0.10	0.45	407	912			-	2,469	**2,407**	2,413	3,112	-	-20.65	-22.65	-22.47	✗	-	✓	✓	✓	✓
5	0.30	0.35	311	896			-	2,385	**2,379**	2,380	2,695	-	-11.51	-11.72	-11.67	✗	-	✓	✓	✓	✓
10	0.01	0.49	485	988	1,469	0.17	-	2,144	**1,607**	1,623	2,824	-	-24.08	-43.08	-42.52	✗	-	✓	✓	✓	✓
10	0.10	0.45	407	912			-	1,850	**1,557**	1,564	2,547	-	-27.36	-38.87	-38.58	✗	-	✓	✓	✓	✓
10	0.30	0.35	311	896			-	1,656	1,515	**1,507**	2,059	-	-19.57	-26.46	-26.82	✗	-	✓	✓	✓	✓
15	0.01	0.49	421	858	1,148	0.16	-	1,830	**1,296**	1,321	2,712	-	-32.51	-52.19	-51.27	✗	-	✓	✓	✓	✓
15	0.10	0.45	335	751			-	1,535	**1,238**	1,250	2,402	-	-36.11	-48.47	-47.97	✗	-	✓	✓	✓	✓
15	0.30	0.35	17	49			-	1,343	**1,186**	1,189	1,482	-	-9.41	-19.97	-19.79	✗	-	✓	✓	✓	✓
Average												-	-18.14	-25.52	-25.30						

(-) no feasible solution; (**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

Table 2: Running times for the ADULT dataset.

					CPU_{tot} [s]					CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	λ	Target	p	q	MIQCP	SetVars	MPFC	Flow	SFC	MPFC	Flow	SFC	MPFC	Flow	MPFC	Flow
2	0.01	0.49	462	941	limit	limit	limit	323.73	limit	44.48	3.24	251.79	-82.33	-98.71	7.16	7.63
2	0.10	0.45	420	941	limit	limit	limit	289.15	limit	41.90	2.89	201.04	-79.16	-98.56	7.08	6.74
2	0.30	0.35	310	893	limit	limit	limit	294.06	limit	38.34	2.94	87.07	-55.97	-96.62	6.78	6.90
5	0.01	0.49	485	988	limit	limit	limit	446.77	limit	164.28	4.47	263.49	-37.65	-98.30	8.41	6.13
5	0.10	0.45	407	912	limit	limit	limit	544.35	limit	138.56	5.44	194.14	-28.63	-97.20	8.62	7.70
5	0.30	0.35	311	896	limit	limit	limit	530.60	limit	120.37	5.31	87.06	38.26	-93.90	8.73	7.53
10	0.01	0.49	485	988	limit	limit	limit	762.23	limit	404.80	7.62	264.43	53.08	-97.12	7.22	6.14
10	0.10	0.45	407	912	limit	limit	limit	1,025.97	limit	327.43	10.26	195.01	67.90	-94.74	8.27	8.67
10	0.30	0.35	311	896	limit	limit	limit	1,142.81	limit	277.83	11.43	87.89	216.11	-87.00	7.85	9.87
15	0.01	0.49	421	858	limit	limit	limit	1,014.20	limit	1,214.26	10.14	231.92	423.57	-95.63	10.00	5.80
15	0.10	0.45	335	751	limit	limit	limit	1,470.32	limit	607.48	14.70	165.75	266.50	-91.13	9.83	8.87
15	0.30	0.35	17	49	limit	limit	limit	1,683.93	1,338.26	515.96	16.84	13.38	3,756.20	25.86	8.57	10.41
Average													378.16	-85.25	8.21	7.70

(limit) time limit of 3,600 seconds reached

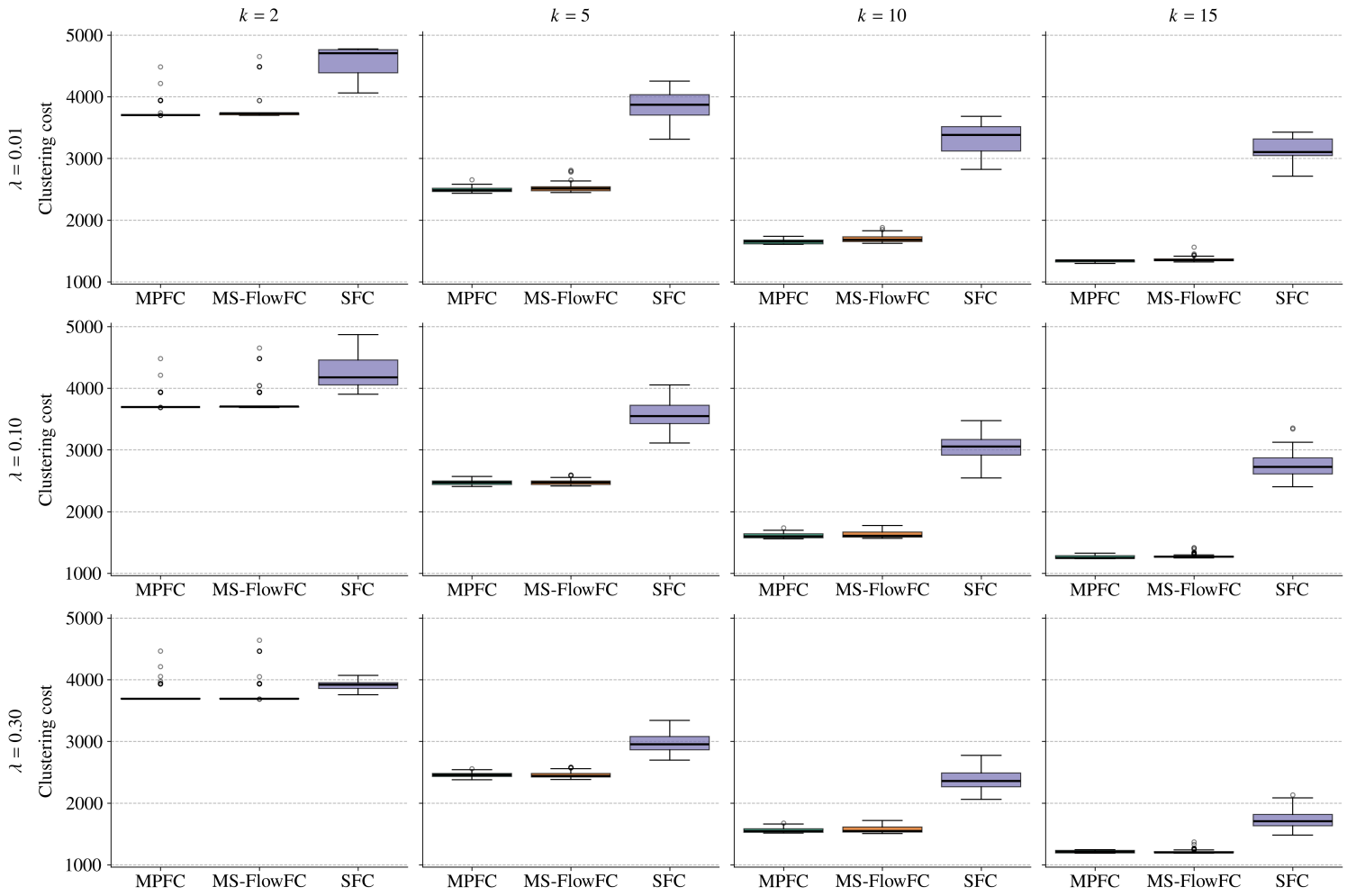

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the ADULT dataset, across different methods, numbers of clusters, and tolerance values.